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Glomerular Diseases Associated With Infection

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More than 250 years ago, dark, scanty urine was observed during convalescence from scarlatina infection. Richard Bright and others identified nephritis symptoms in patients with scarlet fever in the nineteenth century. In the early 1900s, von Pirquet proposed that nephritis occurring after scarlet fever resulted from antibodies that were pathogenic, an insight that unlocked the field of immune-mediated kidney disease. We now recognize that many infectious organisms can induce a spectrum of glomerular lesions via differing pathogenetic mechanisms, including immune complex deposition, direct kidney cell infection and injury, and sequelae of ongoing chronic inflammation (Table 57.1). Glomerular injury may occur secondary to resolved or ongoing infection; therefore, the preferred term is *infection-related glomerulonephritis* (GN).[1] As demographics, health care access, and public health conditions change, there are alterations in the epidemiology, clinical presentation, morphology, and therapeutic approaches to infection-related glomerulonephritides.[2] Table 57.2 shows the glomerular lesions associated with specific microorganisms.

BACTERIAL INFECTIONS

Poststreptococcal Glomerulonephritis

Etiology and Pathogenesis

Poststreptococcal GN (PSGN) is an immune complex–mediated lesion occurring after infection with a nephritogenic bacterial strain. Poststreptococcal GN traditionally was considered a complication of group A β-hemolytic streptococci infections of the skin (M types 49, 55, 57, 60) or upper respiratory tract (M types 1, 2, 4, 12), but PSGN can result from infections caused by other M type streptococci and other groups (particularly group C *Streptococcus zooepidemicus*). There are three immunologic mechanisms likely involved in streptococcal-associated GN: passive entrapment and deposition of circulating immune complexes, in situ immune complex formation associated with nephritogenic planted bacterial antigens or intrinsic antigens displaying molecular mimicry, and deposition of nephritogenic bacterial antigens that activate plasmin resulting in local complement activation.[3-6] Two nephritogenic streptococcal antigens are thought to be involved in PSGN immune complex formation: nephritis-associated plasmin receptor (NAPlr), characterized as glyceraldehyde-3-phosphate dehydrogenase, and the streptococcal pyrogenic exotoxin B (SPEB) together with its more immunogenic zymogen precursor (zSPEB).

Evidence for nephritogenicity includes demonstration of NAPlr and SPEB in kidney biopsies with acute PSGN, and elevation of antibody titers to both antigens in most convalescent sera.[5] In a Japanese study, 92% of PSGN convalescent sera and 60% of sera from patients with uncomplicated streptococcal infections contained anti-NAPlr antibodies.[7] Nephritis-associated plasmin receptor has been localized to glomerular areas with plasmin-like activity but not with complement or immunoglobulin, suggesting local binding to plasmin produces glomerular inflammation and enhances immune complex deposition.[3,7] Streptococcal pyrogenic exotoxin B also can bind plasmin, a possible common mechanism of action. However, unlike NAPlr, SPEB colocalizes with glomerular complement and immunoglobulin G (IgG), suggesting participation in immune-mediated glomerular damage.[4] Streptococcal pyrogenic exotoxin B is the only streptococcal nephritogenic antigen demonstrated in subepithelial "hump" deposits characteristic of PSGN, possibly because of its cationic charge. Studies from Latin America and Central Europe demonstrated only SPEB in PSGN kidney biopsy tissue. In contrast, the *S. zooepidemicus* strain responsible for a Brazilian epidemic lacked the SPEB-related gene, suggesting that different streptococcal antigens induce PSGN in different ethnic groups.[8] Thus, there likely are multiple nephritogenic antigens depending on the streptococcal species and genetic and other host factors.

Poststreptococcal GN is thought to occur when glomerular immune complexes or deposition of plasmin-activating nephritogenic bacterial antigens initiate an inflammatory cascade with local complement activation and recruitment of neutrophils and monocyte-macrophages. Subepithelial immune deposits develop associated with cationic antigens (e.g., SPEB) resulting from subendothelial immune complex dissociation, transit through, and reformation on the outer glomerular basement membrane, or in situ immune complex formation. Complement activation in PSGN has garnered renewed interest. In most patients with PSGN, serum C1q is normal while C3 is reduced, indicating predominantly alternative complement pathway activation. Because streptococcal antigens induce a robust antibody response, the classic complement pathway likely fails to be activated due to streptococcal produced "evasins" that bind C1q *(endopeptidase O)* and degrade IgG (*IdeS/Mac-1, Mac-2,* and *endopeptidase S*). Lectin complement pathway activation, characterized by early C4 level reduction, has been demonstrated in a significant number of patients with PSGN. One study identified mesangial deposits of mannose-binding lectin (MBL) and other lectin pathway components in 7 of 18 patients with PSGN.[9] However, lectin pathway activation is not required, as PSGN can develop in patients with MBL deficiency.

The demonstration of pathogenic autoantibodies in PSGN has renewed interest in other autoimmune aspects of this disease.[10] Rheumatoid factors (especially IgG) and cryoglobulins have been found in 35% of patient sera in the first week of disease. Rheumatoid factors were demonstrated in PSGN kidney tissue and kidney eluates from a fatal case. Loss of sialic acid from autologous IgG, due to streptococcal neuraminidase or from IgG Fc fragment binding to type II Fc receptors in the streptococcal wall, may induce anti-IgG reactivity. This anti-IgG antibody may play a role in deposition of IgG-containing complexes. Additional autoimmune manifestations include anti-C1q antibodies, particularly in severe cases, and rarely anti-DNA reactivity, antineutrophil cytoplasmic autoantibodies (ANCAs), and autoimmune hemolytic anemia.[10]

TABLE 57.1 Pathogenetic Mechanisms of Infection-Related Glomerular Injury

Mechanism	Representative Microorganism	Supportive Evidence
Circulating immune complexes	*Streptococcus*	Circulating antibodies against SPEB and NAPlr antigens
In situ immune complex formation	*Streptococcus*	Identification of SPEB in subepithelial deposits in PSGN
Nephritogenic antigen deposition with complement activation	*Streptococcus*	Glomerular deposition of NAPlr with plasmin activity
Superantigen binding to APCs activates T cells and polyclonal B cells, inducing immunoglobulin production	*Staphylococcus*	Selective peripheral blood activation of TCR V_β repertoire
Direct kidney cell infection	HIV, parvovirus	Virus identification in glomerular epithelial cells
Direct kidney cell injury	HBV	HBV induces mesangial cell proliferation and type IV collagen production in vitro
Cryoglobulin production	HCV	IgM rheumatoid factor against the HCV E2 envelope protein
Complement activation	*Streptococcus*	Low C3 levels, lectin deposits, serum antifactor B autoantibodies

APC, Antigen-presenting cell; *HBV,* hepatitis B virus; *HCV,* hepatitis C virus; *HIV,* human immunodeficiency virus; *IgM,* immunoglobulin M; *NAPlr,* nephritis-associated plasmin receptor; *PSGN,* poststreptococcal glomerulonephritis; *SPEB,* streptococcal pyrogenic exotoxin B.

TABLE 57.2 Infectious Organisms Causing Glomerular Injury

Microorganism	Associated Glomerular Lesions[a]
Bacteria	
Gram Positive	
Streptococci	DPGN, MPGN, DDD, IgA-dominant IRGN
Staphylococci	IgA-dominant IRGN, EPGN, DPGN, MesPGN, ICMGN, crescentic GN
Pneumococcus	EPGN, MPGN
Propionibacterium	MPGN, ICMGN (shunt nephritis)
Corynebacterium	MPGN, ICMGN (shunt nephritis)
Gram Negative	
Brucella	ICMGN
Campylobacter	DPGN, MesPGN
Escherichia coli	IgA-dominant IRGN
Haemophilus	IgAN
Klebsiella	DPGN, IgA-dominant IRGN
Legionella	Crescentic GN, EPGN, MesPGN
Meningococci	MesPGN, DPGN
Pseudomonas	MPGN, ICMGN (shunt nephritis)
Rickettsia	Crescentic GN
Salmonella	MesPGN, IgAN
Serratia	MPGN, ICMGN (shunt nephritis)
Yersinia	IgAN, EPGN
Other	
Bartonella	Crescentic GN (endocarditis)
Coxiella	Crescentic GN (endocarditis)
Leptospira	Segmental glomerular necrosis
Mycobacteria	MPGN, ICMGN, DPGN, amyloidosis
Mycoplasma	ICMGN
Treponema	MN, MPGN, DPGN, MesPGN
Viruses	
Hepatitis A	MPGN, IgAN, MesPGN
Hepatitis B	MN, MPGN + cryoglobins, DPGN with cryoglobulins, EPGN, MesPGN, IgAN, FSGS, amyloidosis
Hepatitis C	MPGN ± cryoglobulins, MN, fibrillary GN, immunotactoid GN, MesPGN, collapsing GP
HIV	Collapsing GP, FSGS, IgAN, ICMGN (including lupus-like), MCD
Parvovirus B19	Collapsing GP, ICMGN, EPGN, IgAN, FSGS
Adenovirus	EPGN, MesPGN
Coxsackie B virus	Proliferative GN
Cytomegalovirus	MPGN, MesPGN, collapsing GP, MN, crescentic GN
Dengue	MesPGN
Epstein-Barr	Collapsing GP, MN, crescentic GN, MPGN, EPGN, MesPGN
Hantavirus	MCD, MesPGN

TABLE 57.2 Infectious Organisms Causing Glomerular Injury—cont'd

Microorganism	Associated Glomerular Lesions[a]
Influenza	MesPGN, DPGN
Measles	DPGN
Mumps	MesPGN, DPGN
Rotavirus	MesPGN
Varicella	MesPGN, DPGN
Parasites and Protozoa (see Table 57.3 for more detail)	
Echinococcus	MN, MPGN
Filaria	MesPGN, MPGN, DPGN ± eosinophils, EPGN, MN, MCD, FSGS, collapsing GP, amyloidosis
Malaria	MPGN, MesPGN, MN, IgAN, MCD, FSGS, collapsing GP
Leishmania	MesPGN, MPGN, amyloidosis
Schistosoma	MPGN, FSGS, MesPGN, MN, amyloidosis
With Salmonella	PIGN
Toxoplasma	MPGN, FSGS, MesPGN
Trichinella	MesPGN, MPGN
Trypanosoma	MesPGN, MPGN

[a]These are glomerular lesions described in humans. More infection-associated glomerulopathies have been identified in animal models.
DDD, Dense deposit disease; *DPGN,* diffuse proliferative glomerulonephritis; *EPGN,* endocapillary proliferative glomerulonephritis; *FSGS,* focal segmental glomerulosclerosis; *GN,* glomerulonephritis; *GP,* glomerulopathy; *ICMGN,* immune complex–mediated glomerulonephritis; *IgA,* immunoglobulin A; *IgAN,* immunoglobulin A nephropathy; *IRGN,* infection-related glomerulonephritis; *MCD,* minimal change disease; *MesPGN,* mesangial proliferative glomerulonephritis; *MN,* membranous nephropathy; *MPGN,* membranoproliferative glomerulonephritis pattern; *PIGN,* immune complex–-mediated postinfectious glomerulonephritis.

Epidemiology

The incidence of PSGN is decreasing worldwide; however, it remains high in low-income countries, with an incidence of 9.3 to 28.5 per 100,000 inhabitants.[2,11] Over the past several decades, likely because of early antibiotic administration, the incidence of PSGN has decreased to 0.2 to 0.6 cases per 100,000 person-years in wealthy countries.[12] Poststreptococcal GN now is more common in older adults with comorbid conditions such as diabetes, alcoholism, and malignancy.[2,6] The higher incidence in low-income countries and in alcoholics likely is associated with poor socioeconomic conditions, limited health care access, delayed antibiotic administration, and a tropical climate.

Poststreptococcal GN is the most common finding in children with acute nephritis in low-income countries occurring sporadically or in epidemics. It typically affects those 2 to 14 years old, with a 2:1 male predominance, and represents a major problem in indigenous populations as in Australia, where more than 95% of cases occur in aboriginal people residing in remote locations.[13] The risk for PSGN varies with infection site and nephritogenic bacterial strain, ranging from 5% in throat infections to 25% in skin infection with the M type 49 streptococcus, and is diminished with early antibiotic therapy. Epidemics have been associated with both skin and throat infection, as well as consumption of unpasteurized milk from cows with mastitis caused by *S. zooepidemicus.* There is also a genetic predisposition: PSGN is associated with human leukocyte antigen (HLA)-DR4 and HLA-DR1 expression, and siblings of sporadically affected patients have a 38% risk for developing clinical or subclinical glomerular disease.[6]

Clinical Manifestations

Poststreptococcal GN presents differently in children and older debilitated adults. Symptomatic children typically present with acute nephritic syndrome, microhematuria in 100%, macroscopic hematuria in 30%, hypertension in 80%, edema in 80% to 90% (chief complaint in 60%), and oliguria in 25% to 40%. Nephrotic syndrome occurs in 2%, and ascites is uncommon. In contrast, 20% of adults with PSGN have nephrotic syndrome, 80% to 86% have hypertension, 83% have kidney impairment, and 43% have congestive heart failure. Rapidly progressive kidney failure with glomerular crescents occurs in less than 1% of children and adults. Asymptomatic subclinical cases have microscopic hematuria, and transient hypocomplementemia is 4 times more frequent compared with symptomatic cases where hypocomplementemia typically is sustained. Serum CH50 is low and reduced C3 levels occur in more than 90% of patients in the first week of disease; these normalize within 2 months of disease resolution. Serum C1 levels are normal and C4 may be depressed early in disease. Eighty percent have elevated IgG and IgM with normal IgA in contrast to rheumatic fever, which has increased IgA levels in 80% of the patients. Cryoglobulins, rheumatoid factor, and anti-C1q antibodies occur in up to one-third of patients in the acute phase. Families may have various manifestations of PSGN; therefore, eliciting a family history for streptococcal infection and signs of glomerular disease is recommended.

Most patients give a history of streptococcal infection, although it often has resolved at presentation. *Streptococcus* cultures are positive in 10% to 70% of epidemic and in 20% to 25% of sporadic cases. The latent period between infection and nephritis symptoms is 2 to 4 weeks after skin infections and 7 to 10 days after throat infections. As cultures are often negative, antistreptococcal antibody levels are used to ascertain prior infection. The most widely used are antistreptolysin O (ASO) titers, increased in more than 65% of patients with PSGN following throat infections, and anti-DNAse B titers, elevated in 73% of postimpetigo cases. The more sensitive streptozyme panel measures anti-DNAse B, antihyaluronidase, antistreptolysin O, and antistreptokinase antibodies, and is positive in more than 80% of patients. Antibody titers to NAPlr and SPEB/zymogen are more sensitive and specific, but not available clinically.[5,11]

Pathology

Kidney biopsy is not routinely indicated in PSGN but may be required for confirmation when clinical features are atypical, such as nephrotic proteinuria, decreased C3 levels for more than a month (suggesting transformation to C3 glomerulopathy), worsening kidney dysfunction, or adult age. The most common appearance is diffuse endocapillary

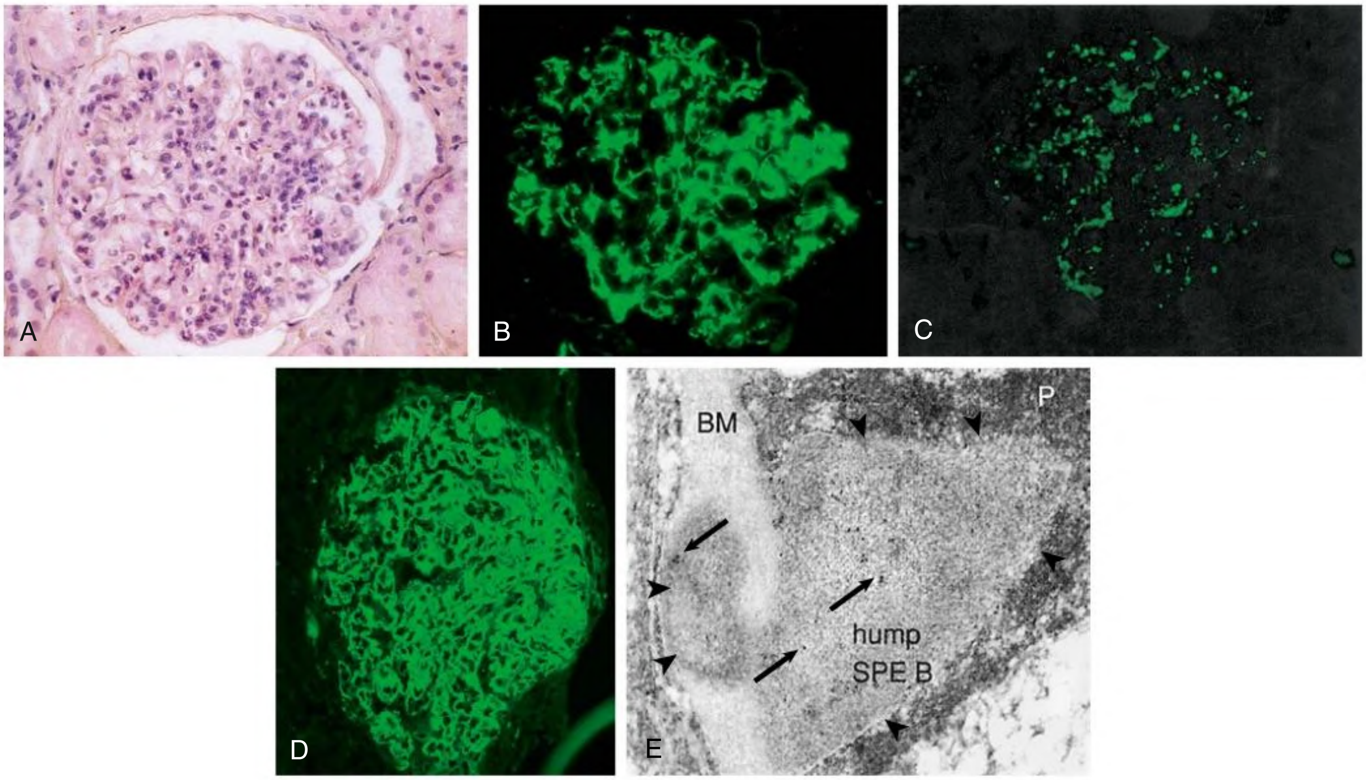

Fig. 57.1 Poststreptococcal Glomerulonephritis. (A) Light microscopic appearance of diffuse proliferative and exudative glomerulonephritis. (B–D) Immunofluorescence for C3 showing the mesangial (B), starry sky (C), and garland (D) patterns. (E) Immune electron microscopy showing the characteristic subepithelial electron dense deposit *(hump)* with intramembranous extension *(arrowheads)*, inside which streptococcal pyrogenic exotoxin B *(SPE B)* is demonstrated *(arrows)* (immunogold staining). *BM,* Basement membrane; *P,* podocyte. (From Rodríguez-Iturbe B, Batsford S. Pathogenesis of poststreptococcal glomerulonephritis a century after Clemens von Pirquet. *Kidney Int.* 2007;71[11]:1094–1104.)

GN characterized by endocapillary neutrophils with fewer monocytes and variable mesangial hypercellularity (Fig. 57.1). Less often or during resolution, there may be diffuse proliferative GN with more mesangial and variable endocapillary hypercellularity, focal endocapillary GN, or mesangial proliferative GN (mesPGN). A membranoproliferative GN (MPGN) pattern is infrequent; however, NAPlr antigen has been identified in biopsy samples with an MPGN pattern, suggesting PSGN may have varied histology. Small numbers of necrotizing and crescentic lesions occur in up to 50% of cases, whereas extensive crescent formation is unusual.[14]

Immunofluorescence shows dominant C3 staining, and the pattern varies with the phase of disease.[15] In acute and subacute disease the immune deposits are in glomerular capillary walls and mesangial regions in a "starry sky" pattern containing strong C3 with less frequent and intense IgG, and IgM in up to 50% of cases; IgA and C1q are minimal or absent. The "garland" pattern is associated with numerous subepithelial hump deposits and heavy proteinuria corresponding to active disease, although also has been described in later disease. As deposits are cleared in subacute to chronic injury, mesangial deposits may persist resulting in a mesangial pattern with predominant C3, loss of immunoglobulins, and fewer infiltrating leukocytes (see Fig. 57.1). Although not routinely assessed, properdin and the terminal membrane attack complex C5b-C9 have been identified in capillary wall and mesangial deposits.[16]

Electron microscopy (EM) demonstrates subepithelial individual hump-shaped deposits, which are typical, although not pathognomonic, of PSGN. These deposits may be infrequent or abundant and are preferentially located where the capillary basement membrane reflects over the mesangium. Glomerulonephritis resolves with deposit clearing and cell apoptosis. However, deposit fragments may remain for years, appearing as weak, irregular, granular glomerular C3 staining with electron-lucent areas or an appearance of striated membranous structures indicative of remote immune complex deposition.[17]

Differential Diagnosis

In a patient with typical symptoms and serologic findings, the diagnosis is straightforward. However, when oliguria lasts more than a week, acute decrease in kidney function lasts 2 weeks, hypocomplementemia lasts more than 1 month, or without a preceding infection history, kidney biopsy is indicated. The differential diagnosis encompasses the spectrum of C3 glomerulopathies and crescentic GN (see Chapter 23). C3 GN may appear morphologically indistinct from PSGN, including subepithelial humps, diffuse endocapillary GN, and weak immunoglobulin deposition. Dense deposit disease demonstrates diagnostic electron-dense transformation of glomerular and tubular basement membranes. Correlating clinical findings, the patient course and C3 levels may be required for a correct diagnosis. Recently, Chauvet and colleagues[18] found that transient antifactor B antibodies, which activate C3 convertase activity, were present in 31 of 34 patients with PSGN and correlated with serum C3 levels. These antifactor B autoantibodies differed from C3 nephritic factor and permitted discrimination between PSGN and C3 glomerulopathy with a sensitivity of 95% and a specificity of 82%. When glomerular crescents on light microscopy occur together with a positive ANCA assay, the differential diagnosis is pauci-immune (ANCA-associated) crescentic GN; the diagnosis of PSGN depends on identifying C3-containing electron-dense deposits in a subepithelial pattern.

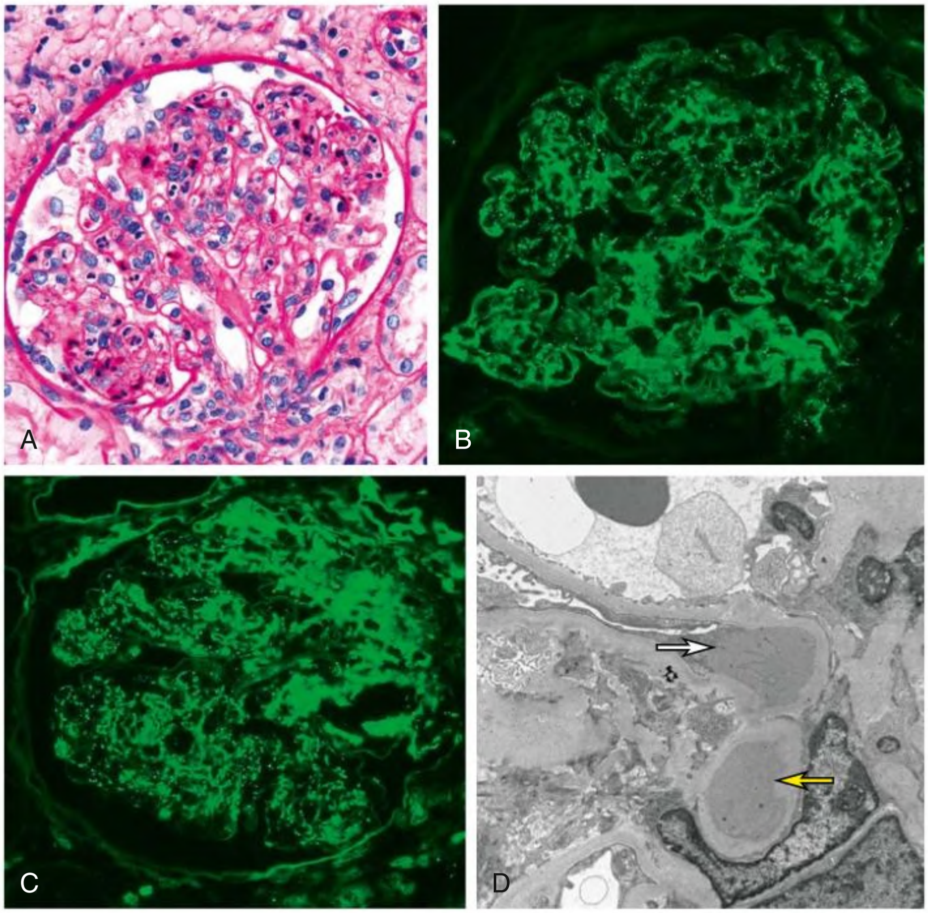

Fig. 57.2 Immunoglobulin A (IgA)-Dominant Glomerulonephritis (GN) in a Diabetic Patient With Methicillin-Sensitive *Staphylococcus aureus* Infection. (A) A diffuse exudative GN is present with neutrophil infiltration. (B) Immunofluorescence shows IgA irregular granular mesangial and capillary wall deposits. (C) Immunofluorescence shows C3 in a pattern similar to that of IgA. (D) Electron microscopy shows dome-shaped subepithelial deposit *(white arrow)* and mesangial deposit *(yellow arrow).*

Natural History

Most children with PSGN recover with increased urine output and clinical symptom resolution in 7 to 10 days, and normalization of serologic tests within 1 month. End-stage kidney disease (ESKD) occurs in less than 1% of children observed for one to two decades after the acute attack.[11] Following recovery, mild proteinuria (<500 mg/day) and microscopic hematuria may persist for up to 1 year without affecting the long-term prognosis. However, older patients with comorbidities are more likely to experience acute complications, including kidney impairment in 30% to 77%, congestive heart failure in 40%, nephrotic proteinuria in 20%, and mortality in up to 30%.[11,19] A nephrotic presentation and persisting proteinuria are associated with worse long-term prognosis. Underlying alternative complement pathway dysregulation may increase risk for progressive kidney disease leading to C3 GN, which is now thought to encompass most cases of “atypical” postinfectious GN. Certain epidemics also have reported a high incidence of chronicity, perhaps related to more affected adults, who tend to have worse outcomes.[20] Studies in Australian aboriginal communities have shown that PSGN in infancy increases risk for subsequent albuminuria, hematuria, and chronic kidney disease (CKD).[19] The latter is attributed to a two-hit injury resulting from imposition of diabetes, metabolic syndrome, or low birth weight on early PSGN.

Treatment

Management includes culture and treatment of any persistent streptococcal infection. Early antibiotic treatment likely prevents PSGN, and prophylactic treatment of family members in high-risk communities or epidemic circumstances may be indicated. Treatment is with oral penicillin, intramuscular penicillin (a single intramuscular injection of 1.2 million units of benzathine penicillin in adults, or half this dose in small children), erythromycin (in patients allergic to penicillin), or cephalosporins. Oral treatments are given in doses every 6 hours for 7 to 10 days.[21] Treatment of acute nephritic syndrome includes fluid and sodium intake restriction and loop diuretic administration. An oral long-acting calcium antagonist is usually sufficient to control hypertension. Intravenous medications may be indicated in exceptional cases with hypertensive emergency. Dialysis (hemodialysis or peritoneal dialysis) is required in 25% to 30% of adults but seldom in children. Poststreptococcal GN complicated with crescents may benefit from pulse methylprednisolone. The prognosis of crescentic PSGN is significantly better than for crescentic GN from other causes, but residual impairment of kidney function occurs in more than half the cases.

Immunoglobulin A–Dominant Infection-Related Glomerulonephritis

Etiology and Pathogenesis

Immunoglobulin A–dominant infection-related GN (IRGN) is an immune complex–mediated lesion typically associated with *Staphylococcus aureus* infections in adult patients with underlying comorbidities.[22] Koyama and colleagues[23] suggested that the pathogenesis involves staphylococcal enterotoxins functioning as superantigens that bind directly to the major histocompatibility complex (MHC) class II molecules on antigen-presenting cells. This enterotoxin–MHC class II complex engages the V_β T cell–receptor region, resulting in T cell activation and a cytokine burst, initiating a B cell polyclonal IgG and IgA response putatively against a *S. aureus* cell envelope antigen.[23] This antigen colocalizes with glomerular IgA deposits and induces mesangial deposits in an experimental model.

Epidemiology

Immunoglobulin A–dominant IRGN usually occurs in the setting of ongoing (rather than resolved) infection in adults, with an average age of 58 and up to 41% over 65 years.[24,25] There is a male predominance, and most patients have comorbidities, including diabetes in up to 65% of patients, malignancy, heart disease, or alcohol or substance abuse.[6] The incidence is unknown, but IgA-dominant IRGN occurred in 1.6% of kidney biopsies in adults in one institution and appears to be increasing.[26,27] In developed countries, it has a 300% increased prevalence compared with PSGN, and many cases of IgA-dominant IRGN likely are asymptomatic and undiagnosed. Approximately 65% of patients are infected with methicillin-resistant *S. aureus* (MRSA), and 25% have methicillin-sensitive *S. aureus* (MSSA).[24] The most frequent sites of infection are the skin and viscera, including pneumonia, endocarditis, and osteomyelitis. Although less common, other pathogens such as *S. epidermidis, Streptococcus, Klebsiella,* and *Escherichia coli* have been associated with IgA-dominant IRGN.[6,22,25]

Clinical Manifestations and Pathology

Active disease often presents with acute or rapidly progressive kidney failure or nephrotic range proteinuria in 50%. Hematuria is universal with gross hematuria in 25%.[24] However, a mild clinical course is not uncommon, with subacute or chronic disease and delayed diagnosis due to underlying chronic disease in elderly patients. Low C3 levels are found in 55% to 70% of patients, and serum IgA levels may be mildly increased.

Kidney biopsy reveals diffuse endocapillary or proliferative GN in 40% to 80% of cases, MesPGN in 20% to 60%, and crescentic GN (≥ 50% crescents) in less than 5%.[24-26] A smaller number of crescents occurs in up to 35% and positive ANCA in 22%.[27] The deposits contain weak to intense IgA with codominant C3 staining in the same pattern and kappa light chain similar to or greater than lambda.[22,27] Electron microscopy shows mesangial deposits in most cases, with up to 83% having variable numbers of subepithelial hump deposits and half with small and less frequent subendothelial deposits[2] (Fig. 57.2). A subset of patients with cirrhosis and staphylococcus, or few other bacterial infections, develop large confluent mesangial and subendothelial (wire loop-like) deposits with fewer subepithelial and intramembranous deposits, typically IgA dominant or codominant.[28]

Differential Diagnosis, Natural History, and Treatment

Differentiation from IgA nephropathy is important for treatment and outcome. Immunoglobulin A–dominant IRGN is favored in the presence of staphylococcal infection, hypocomplementemia, severe proteinuria, and kidney biopsy demonstrating immune staining and electron-dense deposits as detailed earlier and absent IgA in globally sclerotic glomeruli. Recently, differences in kidney tissue protein abundance in IgA nephropathy versus IgA-dominant IRGN were identified by mass spectrometry and confirmed by immunohistochemistry.[29] There is a worse outcome compared with IgA nephropathy, likely because of underlying comorbid conditions. Mesangial IgA deposits may persist despite recovery from infection, possibly related to an underlying risk for IgA immune complex disease.

Active infection should be treated with appropriate antibiotics, and recovery of kidney function may result despite ongoing infection. Corticosteroid treatment is contraindicated, at least during active infection, highlighting the importance of differentiation from IgA nephropathy, in which corticosteroids may be indicated in severe cases (see Chapter 24).

Staphylococcus-Associated Glomerulonephritis

Etiology, Pathogenesis, and Epidemiology

Infection with MRSA, MSSA, and methicillin-resistant and methicillin-sensitive *Staphylococcus epidermidis* can cause GN, usually IgA-dominant IRGN, endocarditis-associated GN, or shunt nephritis. However, a minority of patients with staphylococcal infection will develop GN that does not fit into these categories.[12,24] As described earlier, the infection results in immune complex deposition in most cases. In addition, an *S. aureus* plasmid-encoded peptide homologous to the myeloperoxidase (MPO) T cell epitope has been shown to induce MPO-ANCA, possibly inducing crescents and vasculitis in some patients.[12] Staphylococcal infection–related GN has increased over the past three to four decades, although the incidence and prevalence are unknown. Most patients with staphylococcal-associated GN are adults and have comorbidities.

Clinical Manifestations and Pathology

There is no latent infection-free period, although the presence or source of infection may not be obvious. Low C3 is found in 30% to 50%, C4 levels are normal, and ANCA may be positive.

Hematuria, proteinuria, often in the nephrotic range, and acute kidney injury (AKI) variably occur. Glomerular lesions other than those previously indicated include MesPGN, diffuse endocapillary or proliferative GN, and necrosis and crescents. Immunofluorescence reveals moderate to strong glomerular C3 staining in most biopsies, weak IgA staining in 10%, no IgA or IgG in 15%, and a pauci-immune pattern in 13%.[27] Electron microscopy shows mesangial and few subendothelial deposits, and subepithelial humps in less than one-third of cases.

Differential Diagnosis and Treatment

The differential diagnosis for the manifestations of staphylococcal-associated GNs includes PSGN, IgA nephropathy, IgA vasculitis, C3 GN, ANCA-associated crescentic GN, and rarely cryoglobulinemia. Clinical pathologic correlation and a search for infection, including endocarditis, are necessary. Treatment involves eliminating the infection and supportive care. Immunosuppression is contraindicated during active infection and has not been shown to improve outcome in this setting.

Endocarditis-Associated Glomerulonephritis

Etiology and Pathogenesis

Endocarditis may be an acute or subacute infection. Approximately one-third of patients with endocarditis develop AKI secondary to GN, kidney emboli (infarcts and cortical necrosis), or treatment (drug-induced interstitial nephritis). The pathogenesis of endocarditis-associated GN involves deposition of immune complexes containing bacterial antigens similar to that proposed for PSGN. Less virulent organisms with longer undiagnosed endocarditis may predispose to immune complex formation and development of GN.[30] Polyclonal type III cryoglobulins occur in 50% of patients but rarely deposit in glomeruli. Some bacteria, classically MRSA, express superantigens that can activate T cells directly leading to a polyclonal gammopathy and immune complex GN.

Epidemiology

Historically, infective endocarditis (IE) affected young adults with rheumatic heart disease; however, it now occurs in older and at-risk patients, including drug users, prosthetic valve and implantable device recipients, and those with human immunodeficiency virus (HIV) and hepatitis C infection, diabetes, and health care–associated bacteremia.[30,31] *S. aureus* and *S. epidermidis* cause more than half of the endocarditis-associated

GN[31] and are the most common pathogens in hospital-acquired IE. *Streptococcus* infections are more frequent in community-acquired and native valve endocarditis, although MRSA-associated community-acquired IE is increasing.[32] Gram-negative bacteria (*Enterococcus faecalis, E. coli, Brucella,* and *Proteus*) and HACEK microorganisms (*Haemophilus, Aggregatibacter, Cardiobacterium, Eikenella,* and *Kingella*) less frequently cause IE. Culture-negative IE is usually caused by *Coxiella burnetii, Bartonella* spp., and *Tropheryma whipplei* in untreated patients and by streptococcal species in those with prior antibiotic administration.[33] The HACEK organisms require special laboratory procedures; if these are not used, infected patients will be culture negative. *Bartonella henselae* endocarditis is frequently associated with congenital valvular disease and exposure to cats. Polybacterial infections are usually associated with drug abuse.

Clinical Manifestations

Patients often have fever, arthralgias, anemia, and purpura. Rarely, IE may present as primary kidney disease without systemic symptoms. Classic findings such as Osler nodes, Janeway lesions, and splinter hemorrhages are rare, and the diagnosis may be missed until autopsy in 38% of patients.[30] The most common kidney presentation is AKI, and microhematuria and proteinuria are almost universally present, whereas nephrotic syndrome and rapidly progressive glomerulonephritis (RPGN) are infrequent. Risk factors include intravenous drug in 29% of patients and cardiac valvular disease in 30%.[31] Decreased C3 and C4 levels, elevated rheumatoid factor, circulating immune complexes, and type III cryoglobulins are found in 50%, whereas ANCAs occur in 22% of patients. In those who develop GN, hypocomplementemia, particularly C3, is more frequent, suggesting alternative complement pathway activation. However, complement levels may be normal in superantigen-mediated GN.[31]

Pathology

Autopsy studies have demonstrated glomerular lesions in 7% to 17% of those with IE, including crescentic GN, MPGN, and focal and diffuse proliferative GN patterns, although morphologic details are not uniformly provided.[30] Kidney biopsy findings also are variable, with up to 45% of biopsies demonstrating GN, including 55% crescentic GN, 35% diffuse or focal endocapillary GN, 10% MesPGN, and infrequently MPGN[31] (Fig. 57.3). Immunofluorescence discloses pauci-immune GN in 44%. All proliferative GN cases have C3 staining with IgG or IgA in mesangial regions in more than half and less often in capillary walls. Electron-dense deposits occur in 90% of all GNs, mesangial in 85%, subepithelial humps in 49%, and subendothelial in 45% of the cases.[31,34]

Differential Diagnosis, Natural History, and Treatment

In crescentic GN with or without positive ANCA titers, hypocomplementemia should raise suspicion for underlying IE. Antibiotic treatment usually eradicates the endocarditis and corrects the serologic abnormalities. However, microhematuria, proteinuria, and decreased kidney function may persist for months after successful therapy. The overall mortality of IE is 20% and increases to 36% in patients who develop kidney failure; only 32% of the surviving patients have complete recovery of kidney function.[31] Early diagnosis, risk stratification, and prompt antibiotic therapy are critical for successful resolution, with diminution of circulating immune complex levels correlating with better outcomes.[35] In patients with crescentic GN, pulse plasma exchange with and without steroids has been used in addition to appropriate antibiotic therapy, but the value is unconfirmed. Immunosuppression currently is considered contraindicated in IE-associated crescentic GN.[12]

Shunt Nephritis

Etiology, Pathogenesis, and Epidemiology

Shunt nephritis is an immune complex GN that may develop in association with central nervous system shunt infections. It is rare in ventriculoperitoneal or ventriculovascular shunts presently used, but approximately 6% of previously used ventriculoatrial shunts became infected. Glomerulonephritis develops in 0.7% to 2% with infected shunts 2 months to many years after insertion, and is decreasing due to improved shunt infection rates. The microorganisms are usually *S. epidermidis, S. aureus,* or *Propionibacterium acnes,* and less often diphtheroids, *Pseudomonas, Serratia,* or *Corynebacterium.*

Clinical Manifestations and Pathology

Children are most frequently affected, typically within 6 months of surgery, with recurrent low-grade fever, arthralgias, weight loss, anemia, rash, hepatosplenomegaly, hypertension, and signs of increased intracranial pressure.[12] There may be no signs of systemic infection. Microscopic hematuria occurs in 90%, often with nephrotic range proteinuria. Serologic findings include elevated rheumatoid factor, cryoglobulinemia, decreased serum C3, less often reduced C4, and occasionally positive PR3-ANCA titers. Cerebrospinal fluid may demonstrate eosinophils. Kidney histology shows an MPGN pattern or MesPGN with variable crescent formation, and IgM, IgG, and C3 containing electron-dense deposits in subendothelial and mesangial regions. Cerebrospinal fluid cultures obtained by tapping the shunt are positive for the pathogen, but repeated cultures may be required due to the low numbers of bacteria.

Natural History and Treatment

Treatment requires antibiotic therapy and prompt removal of the infected shunt, usually replaced by a ventriculoperitoneal shunt. Delay in diagnosis and shunt removal worsens the kidney prognosis. If dialysis is required, hemodialysis is preferred because peritonitis carries the risk for meningitis in patients with a ventriculoperitoneal shunt. Complete recovery occurs in more than half of the patients, whereas 22% have persistent urinary abnormalities and 6% develop ESKD.

Glomerulonephritis Associated With Other Bacterial Infections

Osteomyelitis and intraabdominal, pelvic, pleural, and dental abscesses can be associated with GN, typically when infection persists for several months. Kidney disease encompasses mild urinary abnormalities to RPGN, with nephrotic syndrome the most frequent presentation, usually with normal complement levels. Kidney biopsy reveals an MPGN pattern, diffuse endocapillary or proliferative GN, MesPGN, or IgA-dominant IRGN, with or without crescents. If started early, antibiotic treatment may result in recovery of kidney function.

Congenital, secondary, or early latent syphilis may be associated with GN. In congenital syphilis, nephrotic syndrome may be the primary clinical manifestation, with anasarca occurring 4 to 12 weeks after birth. Kidney involvement is uncommon in acquired syphilis, and usually occurs with secondary syphilis. The presentation is nephrotic syndrome or occasionally acute nephritis. Membranous nephropathy (MN) is the most common glomerular finding, but diffuse proliferative GN with or without crescents, an MPGN pattern, MesPGN, and rarely minimal change disease (MCD) have been observed. Treponemal antigens may be identified in the immune deposits. Syphilitic GN responds to antibiotic treatment, often with rapid resolution of nephrotic syndrome in MN, although remission may not occur for 4 to 18 months, particularly with other glomerular manifestations.

Acute typhoid fever from *Salmonella typhi* is characterized by fever, splenomegaly, and gastrointestinal symptoms. In severe cases, disseminated intravascular coagulation or thrombotic microangiopathy

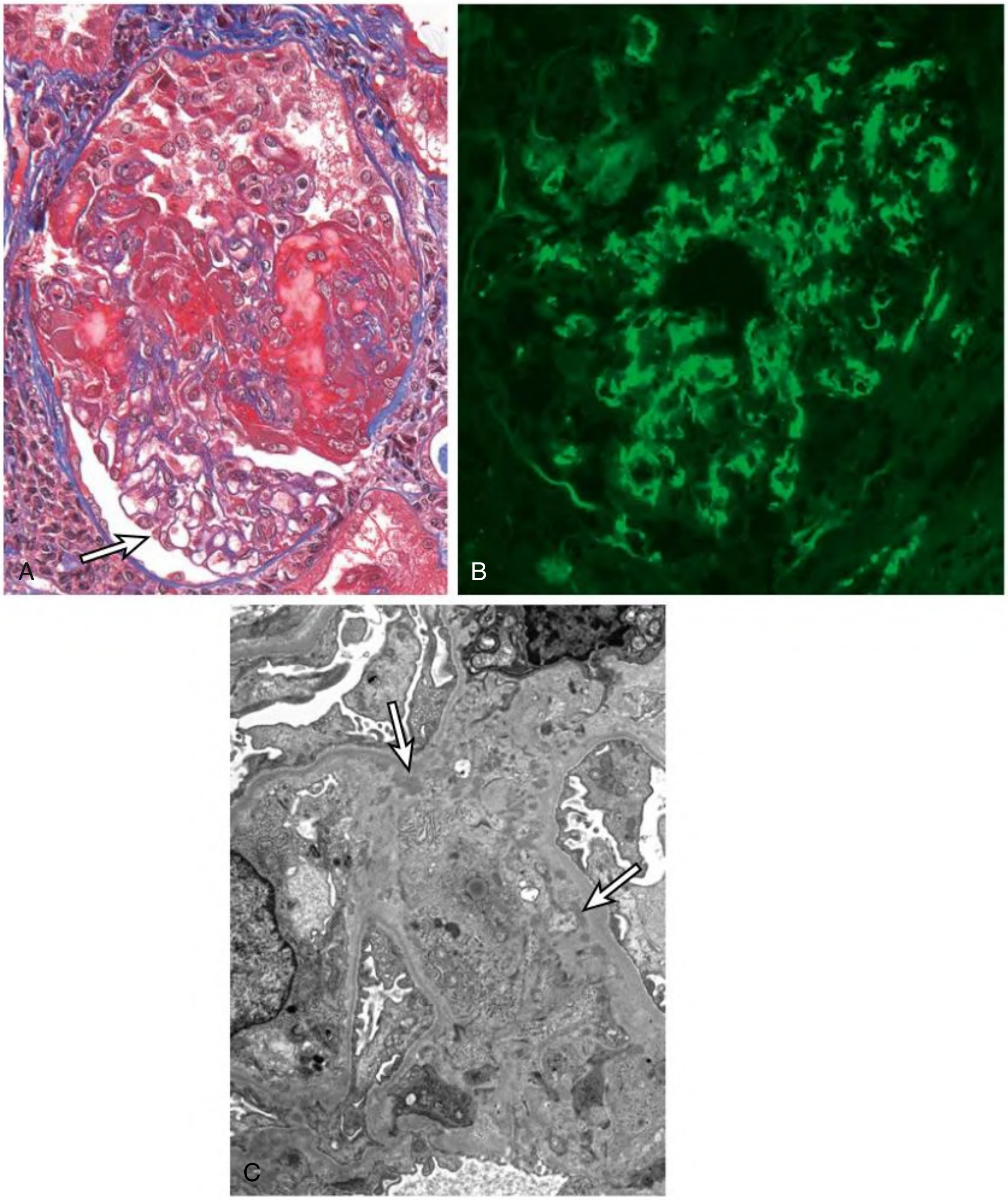

Fig. 57.3 Crescentic Glomerulonephritis Associated With Infective Endocarditis. (A) Glomerulus with a necrotizing and cellular crescent. Note the uninvolved tufts have no inflammation or hypercellularity *(arrow).* (B) Immunofluorescence for C3 with mesangial staining pattern. (C) Small electron dense deposits *(arrows)* are in mesangial regions without subepithelial hump deposits.

occur rarely. Microhematuria and mild proteinuria appear in 25% of patients with MesPGN.[36]

Leprosy (*Mycobacterium leprae* infection) affects 10 to 15 million patients worldwide, with up to 45% having glomerular disease. Glomerulonephritis is immune complex mediated with frequent hypocomplementemia. Nonnephrotic to nephrotic range proteinuria occurs in 2% to 68%, and kidney dysfunction occurs in 4% of patients.[37,38] Nephrotic syndrome is rare and is usually associated with amyloidosis, often with erythema nodosum.[38] Prospective studies demonstrate increased serum creatinine in 35% of patients at some time during their course, and hypertension is common. Kidney biopsy shows proliferative GN and MN with similar frequencies, whereas in autopsy studies, 4% to 31% have amyloidosis, and 5% to 14% demonstrate MPGN or diffuse proliferative GN with IgG, C3, and less frequently IgM, IgA, and fibrin deposition.[36] The incidence of reduced kidney function decreases to 9% after 8 months of multidrug therapy.

Pneumococcal pneumonia is rarely associated with microhematuria and proteinuria, seen with delayed treatment. Diffuse endocapillary GN with crescents and MPGN have been reported, with pneumococcal antigen in the immune deposits. *Streptococcus pneumoniae* is rarely associated with hemolytic uremic syndrome due to unmasking of the glomerular and erythrocyte Thomsen-Friedenreich antigen by pneumococcal neuraminidase A, allowing preformed antibodies to bind and elicit an immune response.

Gastroenteritis caused by *Campylobacter jejuni* may be associated with MesPGN or diffuse proliferative GN. Other bacteria, including *E. coli, Yersinia,* meningococcus, and *Mycoplasma* pneumonia, may induce GN in this setting (see Table 57.2).

VIRAL INFECTIONS

Glomerular injury can occur in several viral infections, primarily hepatitis B virus (HBV), hepatitis C virus (HCV) (see Chapter 22), and HIV (see Chapter 58). These may coincide with acute or chronic infection, depending on the virus and host response, including genetic risk factors such as *APOL1*. The advent of better antiviral therapeutic agents has modified associated glomerulopathies and improved prognoses for patients with virus-associated GN.

Hepatitis A–Associated Glomerulonephritis

Hepatitis A virus (HAV) infection has increased in the last few years in the United States with an epidemiologic shift from contaminated foods to drug abuse and person-to-person transmission leading to updates in vaccination strategies; associated kidney involvement is uncommon.[39] Immunoglobulin A nephropathy has been temporally associated with HAV, with simultaneous resolution of the viral infection and GN suggesting a causal relationship. There are case reports of MesPGN and MPGN, including with dominant IgA. There is experimental evidence for HAV-associated mesangial proliferation and immune complex GN, further supporting causation in humans. Patients present with microscopic hematuria and nonnephrotic proteinuria or may be nephrotic, particularly in association with MPGN. Glomerulonephritis resolution usually coincides with recovery from the viral infection.

Hepatitis B–Associated Glomerular Lesions

Etiology and Pathogenesis

Hepatitis B virus is a DNA virus of the Hepadnaviridae family and contains the hepatitis B surface antigen (HBsAg), hepatitis B core antigen (HBcAg), and hepatitis B e antigen (HBeAg), which is a splice variant of the core antigen. The virus causes immune complex–mediated glomerular injury predominantly through deposition of circulating immune complexes that have been identified in the serum of infected patients with GN. Immune complexes contain the three major hepatitis B antigens, which differ in specific glomerular locations likely related to their size and charge.[1] Hepatitis B e antigen is in smaller cationic subepithelial immune complexes in MN. The larger HBcAg and HBsAg typically deposit in subendothelial and mesangial regions.[1,40] In situ immune complex formation may occur associated with locally expressed HBcAg, evidenced by identification of glomerular HBcAg and its corresponding RNA. Hepatitis B virus may directly injure glomerular cells, as it induces mesangial cell proliferation and increased type IV collagen production in vitro. There likely are contributing host factors, including MHC class II risk alleles.

Epidemiology

Chronic HBV infection is defined as persistence of HBsAg-positive serology without IgM antibodies to HBcAg. Persistent infection occurs in 0.1% to 15% of those with HBV infection and afflicts approximately 350 million people worldwide, who often are asymptomatic. Infection with HBV becomes chronic in more than 90% of infants, 25% to 50% of children infected between 1 and 5 years of age, and 6% to 10% of older children and adults. Vertical transmission (maternal-infant) often occurs in endemic areas, such as China and Southeast Asia. Horizontal transmission follows blood contamination or direct mucous membrane contact. The prevalence of HBV infection is lower in Europe and the United States, with most carriers becoming infected as adolescents or adults by horizontal transmission because of drug abuse, blood transfusions, or sexual relations.

Clinical Manifestations and Pathology

Acute HBV infection may cause nausea, vomiting, fever, hepatomegaly, and a short-lived serum sickness–like syndrome with urticaria, maculopapular rash, neuropathy, arthralgias, arthritis, microscopic hematuria, and nonnephrotic proteinuria. There are no published kidney biopsy studies in acute infection, and kidney disease associated with acute infection resolves in 1 to 2 months associated with viral illness recovery. In chronic HBV infection, 3% to 5% of patients develop kidney disease; additionally, 10% to 30% of patients with chronic HBV are coinfected with HCV and 5% to 10% with HIV.[1,40]

Hepatitis B Virus–Associated Membranous Nephropathy

Membranous nephropathy is the most common glomerular disease in chronic HBV carriers. Children usually present between 6 and 12 years of age and show a strong male predominance, asymptomatic proteinuria or nephrotic syndrome, microhematuria, normal kidney function, and minimal liver disease.[40] The prognosis is usually good with frequent spontaneous remission associated with anti-HBeAg antibody production. Adults develop proteinuria or nephrotic syndrome, often with impaired kidney function and liver disease; approximately 30% will progress to CKD and 10% to ESKD. Hypocomplementemia occurs in less than half the patients.

Kidney biopsy shows MN with active to clearing deposits depending on lesion chronicity. In contrast to primary MN, there often is mesangial hypercellularity with mesangial and/or subendothelial deposits indicative of a form of secondary MN. Endocapillary hypercellularity and endothelial cell tubuloreticular inclusions may occur. Circulating antiphospholipase A2 receptor (PLA2R) antibodies and PLA2R in subepithelial deposits typically are associated with primary MN. However, in HBV-endemic areas in Asia, PLA2R has been reported to colocalize with HBV antigens in glomerular deposits in HBV-associated MN; the significance of this is uncertain.[1] Immunoglobulin G subclass staining of deposits may be helpful, as IgG4 usually dominates in primary MN but is nondominant in secondary MN.[40] Both HBeAg and HBsAg may be identified in glomerular deposits, and HBeAg has been found in eluted proteins (Fig. 57.4).

Hepatitis B Virus–Associated Glomerulonephritis With a Membranoproliferative Pattern

An MPGN pattern is the second most frequent HBV-associated glomerular lesion, typically found in adults. Patients present with nephrotic or less frequently nonnephrotic proteinuria and microhematuria; half are hypertensive, and 20% have reduced kidney function.[1] This pattern is associated with serum HBsAg and HBcAg and reduced serum C3 and C4 levels. Cryoglobulinemia, type II or more often type III, occasionally occurs in HBV-infected patients, whereas type II is more prevalent with concurrent HCV infection.[1,40] Kidney biopsy shows an MPGN pattern with additional subepithelial deposits if there is concomitant MN, and HBsAg may be identified in mesangial and subendothelial deposits (see Fig. 57.5).

Other Hepatitis B Virus–Associated Glomerular Lesions

There are several reports of IgA nephropathy associated with HBV infection, including some that have entered a sustained remission of kidney disease following successful HBV treatment with pegylated interferon. Other studies have shown similar outcomes for patients with IgA nephropathy with and without HBV infection, possibly due to a coincidental association rather than a causal relationship. Immunoglobulin A nephropathy is highly prevalent in HBV-endemic areas in Asia, and mesangial IgA deposits occur in chronic liver disease secondary to impaired clearance of IgA circulating immune complexes (see Chapter 24). There is a report of 10 patients with chronic HBV infection and all three HBV antigens in the serum, showing diffuse endocapillary GN with uncommon subepithelial humps and glomerular HBsAg and HBV DNA.[41] Amyloid A amyloidosis is associated with chronic inflammatory diseases, has been identified in adults and children with chronic HBV infection, and presents with proteinuria

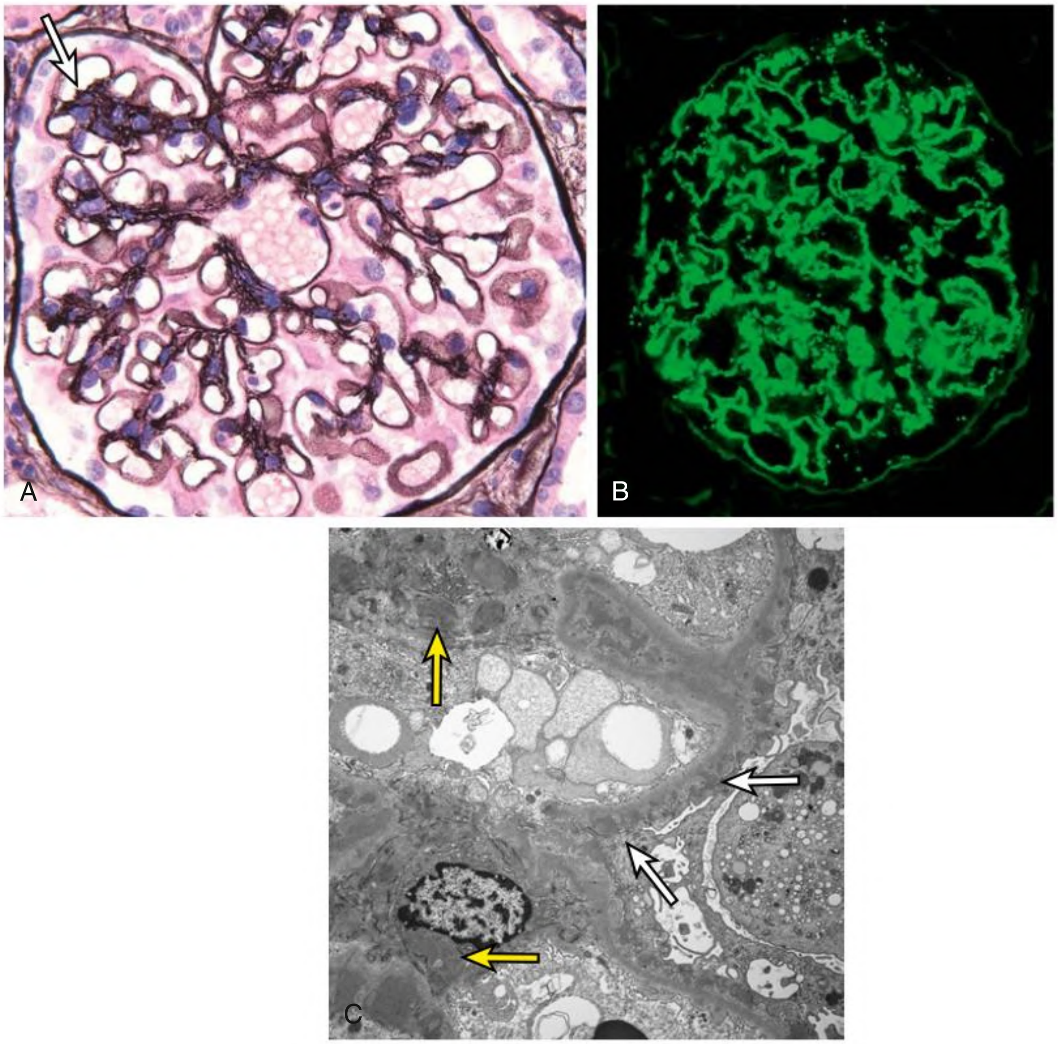

Fig. 57.4 Hepatitis B–associated membranous nephropathy. (A) Irregular glomerular capillary due to small subepithelial spikes with segmental mesangial hypercellularity *(arrow).* (B) Immunofluorescence shows granular immunoglobulin G along capillary walls and segmentally in mesangial regions. (C) Electron microscopy shows subepithelial *(white arrows)* and mesangial *(yellow arrows)* electron-dense deposits.

or nephrotic syndrome. There are rare reports of MCD and collapsing glomerulopathy associated with HBV infection, with HBV DNA in shed urinary podocytes in the latter, suggesting direct HBV infection. Clinical remission occurred after successful HBV infection treatment.[40]

Polyarteritis Nodosa

This vasculitis has been associated with HBV infection and is discussed in Chapter 26.

Treatment

This section addresses only the treatment of patients with GN related to HBV infection (positive HBe antigen or HBV DNA more than 2000 units in HBeAg negative patients). Antiviral treatment is indicated in this setting, as there is consistent evidence that treatment response is associated with reduction in proteinuria. Interferon or nucleoside/nucleotide monotherapy induces seroconversion of positive HBeAg to anti-HBe, and the decrease in HBV DNA levels is associated with significant reduction of proteinuria in 30% to 80% of patients.[42,43]

Interferon therapy has a high rate of seroconversion and sustained remission, although side effects can be frequent and severe.[44] Pegylated interferon may be used in young patients with mild hepatitis and low viral load if not contraindicated (advanced liver disease, leukopenia, thrombocytopenia, depression, pregnancy) but should not be used to treat patients with replicative HBV infection and glomerulonephritis.

In most patients, nucleoside/nucleotide therapy may be chosen, including lamivudine, adefovir, telbivudine, entecavir, or tenofovir, which are all given orally. Nucleoside/nucleotide analog administration should be maintained until HBe antigenemia disappears, usually 4 to 5 years or longer. Early therapy discontinuation increases the risk of virus-related liver failure. All nucleoside/nucleotide drugs are excreted by the kidney in unchanged form, and some have dose-related nephrotoxicity,[44] which is increased with comorbidities including diabetes, hypertension, and CKD.[45]

Entecavir and tenofovir are preferred initial therapies due to higher antiviral activity and lower resistance rates compared with other nucleoside/nucleotide agents. *Entecavir* is generally safe and is used for treatment and prophylaxis of HBV infection in kidney transplant patients. The dose of entecavir is 0.5 mg/day when CrCl is 50 mL/min or more, 0.25 mg/day or 0.5 mg every 48 hours if CrCl is 30 to 50 mL/min, 0.15 mg/day or 0.5 mg every 72 hours if CrCl is 10 to 29 mL/min, and 0.5 mg weekly if CrCl is less than 10 mL/min or the patient is on hemodialysis. Entecavir treatment can be complicated by lactic

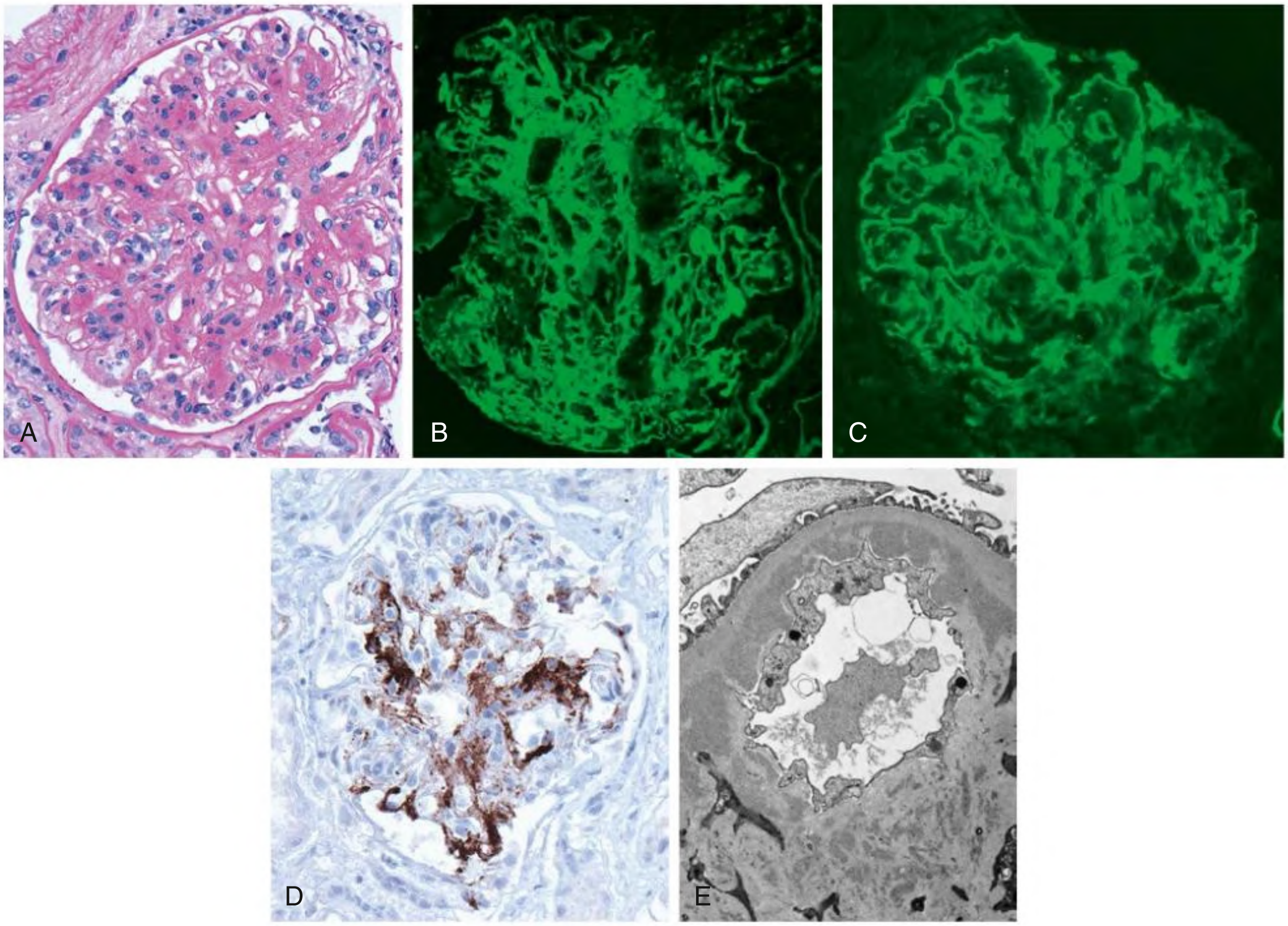

Fig. 57.5 Hepatitis B–Associated Membranoproliferative Glomerulonephritis (MPGN). (A) MPGN pattern of injury with a lobular hypercellular glomerulus. (B) Predominantly mesangial staining for C3 by immunofluorescence. (C) Capillary wall and lesser mesangial staining by immunofluorescence for immunoglobulin M. (D) Hepatitis B surface antigen staining most prominent in mesangial regions. (E) Subendothelial and mesangial electron dense deposits.

acidosis in patients with impaired liver function. *Tenofovir* is used as initial therapy or in patients with resistance to other drugs. The main adverse effects are reduced bone density (osteomalacia) and nephrotoxicity with proximal tubular mitochondriopathy and Fanconi syndrome. Formulations include tenofovir disoproxil fumarate (TDF) and tenofovir alafenamide (TAF), the latter a newer formulation with better stability and more hepatic uptake, requiring substantially lower doses for comparable antiviral activity. A dose of 25 mg of TAF induces similar reduction of HBV DNA levels as low as 300 mg of TDF without adverse effects.[46] Patients switched from TDF to TAF demonstrate improvements in bone density and urinary biomarkers. The dose of TDF is 300 mg/day if CrCl is 50 mL/min or higher, 300 mg every 48 hours if CrCl is 30 to 49 mL/min, 300 mg every 72 to 96 hours if CrCl is 10 to 39 mL/min, and 300 mg once a week if the patient is on hemodialysis. The dose of TAF is 25 mg/day if CrCl is 15 mL/min or more, and in those with CrCl less than 15 mL/min and on hemodialysis, the dose is 25 mg after dialysis on hemodialysis days. Insufficient data are available to recommend doses of TDF or TAF in patients with CrCl less than 10 to 15 mL/min who are not receiving hemodialysis or who are receiving peritoneal dialysis.

Other nucleoside/nucleotide drugs are less effective with a higher incidence of resistance. *Lamivudine* is a second-line therapy with a 20% annual rate of resistance. Lamivudine dosage (100 mg/day) is reduced to 50 mg/day after the first dose if CrCl is 30 to 49 mL/min, to 25 mg/day if CrCl is 15 to 29 mL/min, to 15 mg/day if CrCl is 5 to 14 mL/min, and to 10 mg/day in those on hemodialysis or peritoneal dialysis. Serious adverse effects are rare and include liver failure and rhabdomyolysis. *Telbivudine* has a 25% annual resistance rate, and serious adverse effects include myopathy (cumulative incidence of creatine kinase elevation is 84%), peripheral neuropathy (particularly combined with peg-interferon), hepatoxicity, and lactic acidosis. *Adefovir* is rarely used today due to high resistance rate and dose-dependent nephrotoxicity and Fanconi syndrome that may persist for months after drug withdrawal. Antiviral drug combinations have been used with variable success, but there is no convincing evidence that initial combination therapy with two nucleoside analogs is better than monotherapy.

Patients with concomitant HBV and HCV infections and GN should have HBV DNA and HCV RNA assessed to determine which virus is more likely responsible for the kidney disease. Direct-acting antivirals used to treat HCV infections may activate HBsAg without affecting HBV-associated GN; therefore, patients with combined infections should be treated with interferon that is active against both viruses or have concurrent administration of direct-acting antivirals and nucleoside/nucleotide drugs.

Treatment of HBV-related GN should be limited to antiviral therapy in most patients because seroconversion from HBeAg positive to anti-HBe and reduced HBV DNA levels usually induce proteinuria remission, whereas immunosuppression may stimulate viral replication.

To be noted, a retrospective analysis of 317 patients with HBV-related GN (12 clinical trials) who received combined antiviral and immunosuppressive therapy showed the majority of patients had reduced proteinuria without increase in HBV-DNA viral replication or alterations in liver function.[47] However, immunosuppressive agents are best avoided in patients with replicative HBV infection. In patients with simultaneous HBV infection and anti-PLA2R antibody-mediated MN, treatment with rituximab and cyclophosphamide should be postponed until a sustained virologic remission has been obtained by nucleoside or nucleotide analog therapy.

In specific settings, such as RPGN or severe polyarteritis nodosa (PAN), combination therapy appears to be useful (see Chapter 26). Whenever combined antiviral and immunosuppressive therapy is used, it is important to monitor HBV DNA levels for at least 6 months after immunosuppressive therapy is discontinued.

Other Virus-Associated Kidney Disease

Cytomegalovirus (CMV) infection rarely causes diffuse proliferative GN with immune deposits containing CMV antigens in native kidneys. Cytomegalovirus also has been linked to collapsing glomerulopathy and ESKD in patients without HIV infection. In kidney allograft patients with CMV infection, immunosuppression is reduced (see Chapter 110). Depending on infection severity, oral valganciclovir 900 mg twice daily or ganciclovir 5 mg/kg intravenously every 12 hours is administered until clinical recovery and two blood samples negative for CMV by polymerase chain reaction are obtained, followed by 3 months of oral valganciclovir 900 mg/day or oral ganciclovir 1 g three times daily. Toxic effects of these drugs include neutropenia, central nervous system dysfunction, and carcinogenicity and teratogenicity in experimental studies.

Parvovirus B19 is a single-stranded DNA virus with marked tropism for erythroid precursor cells. Its receptors include the erythrocyte P antigen and glycosphingolipid globoside (Gb4). Infection with a viral prodrome may be followed by nephritic or nephrotic syndrome; nephrotic syndrome may precede an aplastic crisis in patients with sickle cell disease. In transplanted patients, parvovirus infection should be suspected when there is severe anemia resistant to erythropoietin treatment with inappropriate reticulocyte response. Diffuse endocapillary and proliferative GN and MPGN have been reported with subendothelial and fewer mesangial IgG and C3 deposits.[48] Collapsing glomerulopathy has been attributed to parvovirus B19 infection; some, but not all, studies have detected viral DNA in kidney biopsy tissue, and there appears to be an association with *APOL1* risk alleles. There is no specific antiviral therapy, and treatment is intravenous immunoglobulin administration 2 g/kg given over a period of 5 days.

Dengue hemorrhagic fever is currently the most prevalent mosquito-borne urban viral infection worldwide, and is caused by four serotypes of the Flaviviridae virus family. The most common kidney complication is AKI (see Chapter 71). Glomerular involvement may be secondary to direct glomerular cell injury or immune complex deposition. Infected patients may present with hematuria, proteinuria, and kidney failure associated with diffuse endocapillary or proliferative GN with IgG, IgM, and C3, transient IgA deposits, or MPGN.[49] There are no specific treatments, and management is supportive.

Mild kidney abnormalities occur in acute Epstein-Barr virus (EBV) infection, with microhematuria and proteinuria in 10% to 15% of cases associated with diffuse endocapillary GN, MesPGN, MPGN pattern, and collapsing glomerulopathy. Many viral infections, including varicella, mumps, adenovirus, coxsackievirus, and influenza, can be associated with transient microscopic hematuria, nonnephrotic proteinuria, and a mesangial or diffuse proliferative GN in which viral antigens can be identified in the mesangium. Measles may cause diffuse proliferative GN but is better known for its unique ability to induce remission in patients with MCD and nephrotic syndrome as infection abates, due to increased production of Treg cells.

PARASITIC INFECTIONS

Kidney involvement is common in various parasitic infections and includes a range of kidney lesions (Table 57.3). With the exception of malaria, leishmaniasis, filariasis, and schistosomiasis, glomerular involvement is usually mild and transient.[50,51]

Malaria

Etiology and Pathogenesis

There are four major species of *Plasmodium,* two of which typically are associated with GNs in humans: *Plasmodium falciparum* and *Plasmodium malariae.*[52] Malarial antigens activate the immune system by interaction with monocytes, secretion of chemokines and cytokines, polyclonal B cell activation, and circulating immune complex formation. Infected erythrocytes may participate in immune activation through preferential expression of surface antigens and endothelial activation. Malarial antigens have been identified in 25% of glomerular immune complexes and in glomeruli without immune complex deposition, suggesting in situ immune complex formation. Additionally, the alternative complement pathway may be directly activated.[36,52] Concomitant infection with other parasites (schistosomiasis) or viruses (EBV, HIV), or genetic factors such as *APOL1* risk alleles may participate in the pathogenesis of kidney injury and influence geographic disease variations.

Epidemiology, Clinical Manifestations, and Pathology

Malaria is the most prevalent endemic global infectious disease, with 200 million cases predominantly occurring in Africa, India, Southeast Asia, and Latin America causing 600,000 deaths each year (Fig. 57.6).[53] It is acquired by the bite of infected *Anopheles* mosquitoes or infected vectors from blood transfusion or infected organ transplants. *P. falciparum* invades erythrocytes of any age and may cause AKI and multiple organ failure (see Chapter 71). Transient mild proteinuria and microhematuria occur in 25% to 50% of patients, more often children, with infrequent nephritic syndrome and impaired kidney function. Patients develop MesPGN with deposits of IgG, IgM, and C3 and rarely IgAN (Fig. 57.7). Patients present with nephrotic syndrome due to focal segmental glomerulosclerosis (FSGS), infrequently MCD, and most often collapsing glomerulopathy with and without concomitant HIV infection, and associated with African descent and *APOL1* risk alleles.[54] In contrast, *P. malariae* (quartan malaria) infects aging erythrocytes and is less severe. Patients present with steroid-resistant nephrotic syndrome (tropical nephritis or nephrotic syndrome) primarily in children 4 to 8 years old. Adult GN often manifests with hypertension and reduced kidney function. Kidney biopsy shows an MPGN pattern with granular IgG, IgM, and C3 deposition and subendothelial and mesangial electron-dense deposits, and infrequently MN or MCD[52,55] (Fig. 57.8). Despite treating the infection, with or without corticosteroids, there is progression to ESKD, which is attributed to continuing autoimmune mechanisms. *Plasmodium vivax* causes AKI and rarely thrombotic microangiopathy (see Chapter 71), but GN has not been reported. For severe malaria the treatment is artesunate intravenously 2.4 mg/kg per dose at hours 0, 12, and 24, then artemisinin-based combination therapy orally for 3 days. In uncomplicated malaria, recommended treatment options include Artemether-lumefantrine (1.4–4.4/10–16 mg/kg) by mouth twice daily for 3 days or artesunate (4 mg/kg) plus amodiaquine (10 mg/kg) orally daily for 3 days or artesunate (2 mg/kg) orally daily plus tetracycline (4 mg/kg) orally every 6 hours for 7 days. Immunosuppressive treatment is not indicated for malarial nephropathy.

TABLE 57.3 Kidney Lesions in Parasitic Infections

Parasite	Glomerular Lesions	Tubulointerstitial Lesions	Amyloidosis	Tissue Reaction, Symptoms	Acute Kidney Injury	Posttransplant Disease
Schistosomiasis						
S. haematobium[a]	MesPGN DPGN with *Salmonella*	+++	++	Granulomas		+
S. mansoni[a]	MesPGN, MPGN, FSGS, MN, DPGN with *Salmonella*	+	++	Granulomas		+
S. japonicum	MesPGN, MPGN (primates, rabbits)	+				
S. mekongi		+++				
Malaria						
Plasmodium malariae[a]	MPGN, MesPGN, MCD (rare), FSGS (rare), MN (rare)					
Plasmodium falciparum[a]	MesPGN, DPGN, collapsing GP, FSGS, MCD (rare), MPGN (rare), IgAN (rare)	+			++	+
Filariasis						
Onchocerciasis[a]	MesPGN, MPGN, MCD			River blindness		+
Loiasis	MN, MesPGN, MPGN, FSGS (rare)					
Bancroftiasis	MesPGN, MPGN, EPGN, collapsing GP		+	Chyluria Pneumonia Elephantiasis		
Dirofilariasis	MPGN (dogs), MN (dogs, cats)					
Other						
Brugia malayi	MesPGN, MPGN, DPGN	+ (rare)	+ (rare)			
Visceral leishmania	DPGN, MesPGN, MPGN	++	++		+	+
Kala-azar[a]						
Trichinosis	MesPGN, MPGN				+	
Strongyloidiasis	MesPGN					
Echinococcosis	MPGN, MN (rare)			Hydatid cysts		
Opisthorchiasis[a]	MesPGN	++	+		++	
Chagas disease[a]	MesPGN (mice)					
Babesiosis	MesPGN (rat)				++	+
Trypanosomiasis	MesPGN, MPGN (monkey, rat)					
Toxoplasmosis[a]	MesPGN, MPGN, FSGS (rare)	+				

[a]Parasitic antigens or specific antibodies detected in the glomeruli.
All conditions documented in humans unless otherwise indicated.
DPGN, Diffuse proliferative glomerulonephritis; *EPGN,* endocapillary proliferative glomerulonephritis; *FSGS,* focal segmental glomerulosclerosis; *GP,* glomerulopathy; *IgAN,* immunoglobulin A nephropathy; *MCD,* minimal change disease; *MesPGN,* mesangial proliferative glomerulonephritis; *MN,* membranous nephropathy; *MPGN,* membranoproliferative glomerulonephritis pattern.

Malaria-associated kidney disease in low-income countries has declined due to malarial control and overall improved health care.

Filariasis

Filariae are nematodes, of which only eight of the hundreds of species affect humans and only three are associated with kidney disease. *Wuchereria bancrofti* is responsible for several clinical syndromes, including tropical eosinophilic pneumonia, elephantiasis, and chyluria, and induces proteinuria, hypoproteinemia, and hematuria in infected patients. The dominant glomerular lesion is MesPGN; less frequently an MPGN pattern or acute endocapillary GN with eosinophilic predominance can be seen; and rarely amyloidosis or collapsing glomerulopathy are found. Immune complex deposits containing worm antigens have been demonstrated in glomeruli.[52,56]

Bancroftiasis is often associated with bacterial infection, usually staphylococci and certain streptococci; it is uncertain if this contributes to the development of GN. Even less understood is a potential role of the rickettsia-like bacteria *Wolbachia,* discovered within filarial nematodes and blamed for activation of innate immune pathways. Ivermectin, diethylcarbamazine, and albendazole are approved treatments for filariasis.

Onchocerca volvulus is less common and causes scrotal lymph node obstruction ("hanging scrotum"). Onchocercosis is associated with immune-mediated manifestations, including keratitis, anterior uveitis, and optic atrophy leading to "river blindness." GN induces proteinuria, including steroid-resistant nephrotic syndrome with progressive kidney impairment, usually an MPGN pattern with IgM, IgG, and C3 in subendothelial and mesangial deposits, with less frequent MCD and sclerosing glomerular lesions.[52,56] Glomeruli may contain onchocercal antigens, and GN recurs in transplanted kidneys. When *W. bancrofti* or *O. volvulus* is treated, the proteinuria and hematuria may worsen due to parasitic disintegration and increase in circulating immune complex formation. After successful parasite eradication, the kidney disease typically resolves.

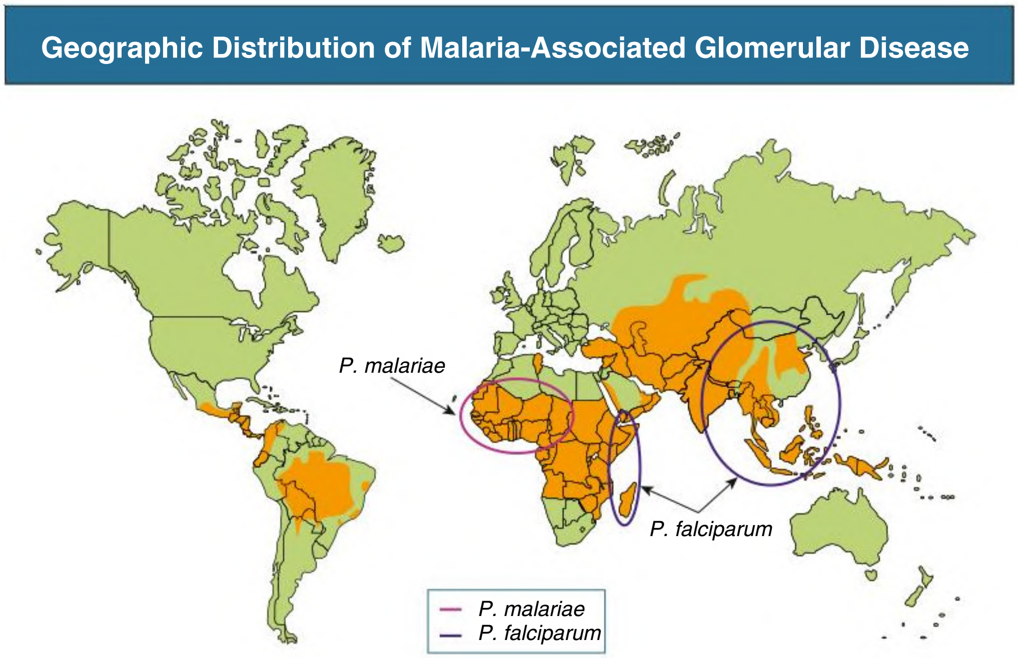

Fig. 57.6 Geographic Distribution of Malaria-Associated Glomerular Disease. Although malaria is endogenous to many areas of the world *(shaded orange),* the major areas where malaria-associated glomerular disease has been reported *(ringed)* and their respective species are shown.

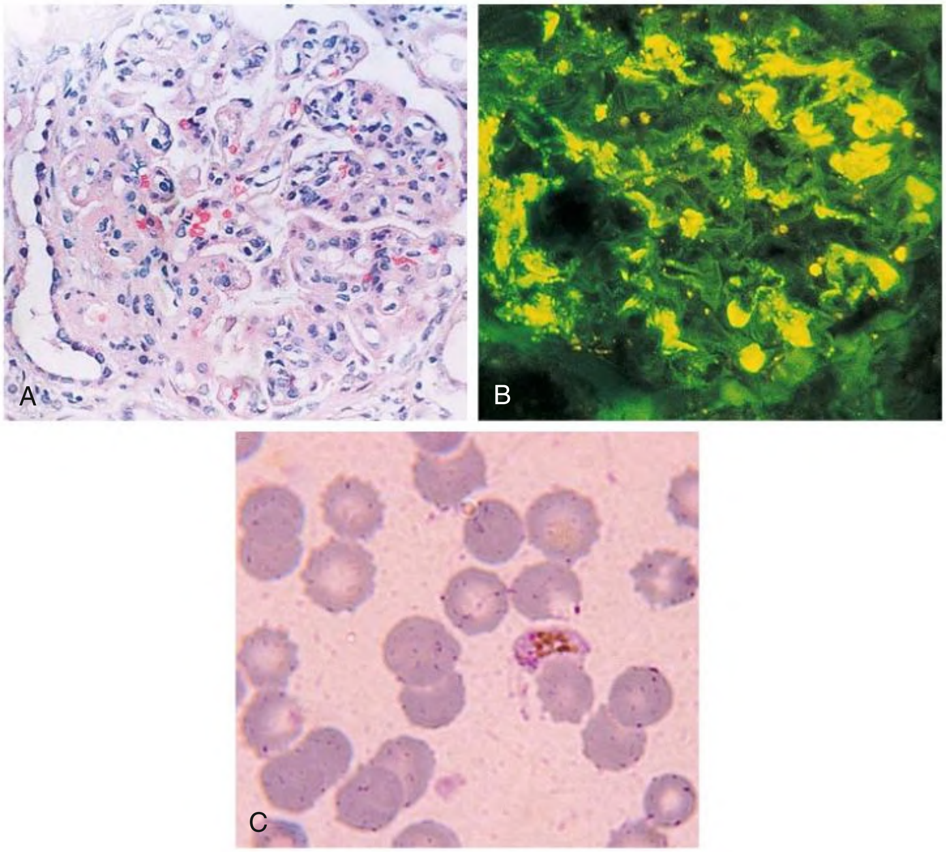

Fig. 57.7 Glomerulonephritis Associated With *Plasmodium falciparum* Malaria. (A) Light microscopy shows mesangial proliferative glomerulonephritis. (B) Immunofluorescence may reveal *P. falciparum* antigens in a mesangial pattern. (C) Peripheral blood smear confirms acute *P. falciparum* infection, with banana-shaped gametocytes and multiple ring forms in erythrocytes. (A, From Barsoum RS. Malarial nephropathies. *Nephrol Dial Transplant.* 1998;13[6]:1588–1597. B and C, Courtesy V. Boonpucknavig.)

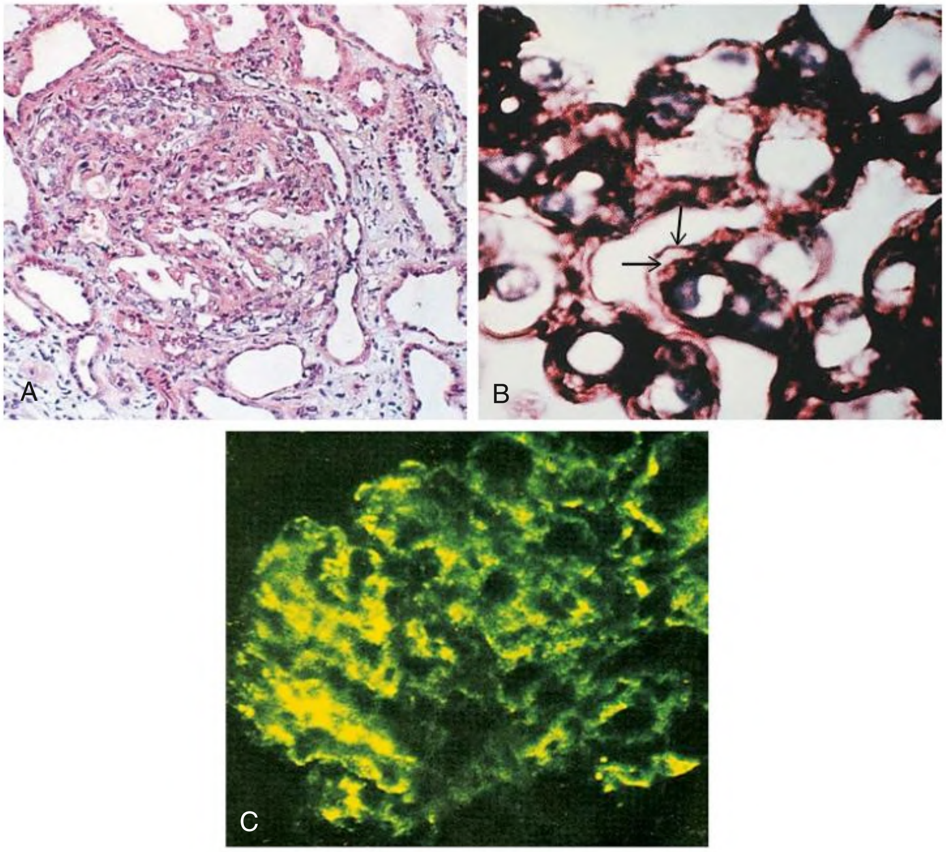

Fig. 57.8 Quartan Malarial Nephropathy. (A) Light microscopy shows sclerosing membranoproliferative glomerulonephritis. (B) Silver stain shows a double contour *(arrows)* of the basement membrane. (C) Malarial antigens are present on immunofluorescence. (From Barsoum RS. Malarial nephropathies. *Nephrol Dial Transplant.* 1998;13[6]:1588–1597.)

Loa is the least prevalent of the nephritogenic filariae. Associated glomerular disease includes MN and less commonly MPGN and FSGS.[52,57] Unlike the other forms of filariasis, treatment for *Loa loa* generally does not result in GN resolution. All three filaria parasites often occur with HBV and/or malaria coinfection, which independently can cause GN; it is unknown whether this affects the incidence or types of GN associated with filariasis.

Leishmaniasis

Leishmania are obligatory intracellular protozoa that infect 12 million humans and many animals. Of 30 known leishmanial spp., only visceral leishmaniasis (kala-azar) induced by *Leishmania donovani* or by the closely related *Leishmania infantum (Leishmania chagasi)* involves the kidney. Sixty percent of those infected with *L. donovani* have mild proteinuria and microscopic hematuria. There is glomerular immune complex deposition with MesPGN or an MPGN pattern; the latter may occur with cryoglobulinemia.[58,59] Amyloidosis may occur, usually associated with coincident HIV infection. Treatments for leishmaniasis (pentavalent antimony compounds, liposomal amphotericin B, miltefosine) have substantial adverse effects and high rates of resistance.

Schistosomiasis

Schistosomiasis is a common cause worldwide of glomerular disease and is discussed in Chapter 56.

SELF-ASSESSMENT QUESTIONS

1. At the present time, infection-related GN in wealthy countries is associated with debilitating conditions (neoplasia, diabetes, AIDS, alcoholism) and IgA deposits.
   A. True
   B. False
2. Patients with acute poststreptococcal GN usually have:
   A. low serum CH50 and C3 levels.
   B. low serum C4 levels.
   C. high serum C3 levels.
   D. high serum CH50 and high serum C4 levels.
3. Patients with infective endocarditis may have crescentic GN without immune deposits.
   A. True
   B. False
4. Combined antiviral therapy and immunosuppression (cyclophosphamide and steroids) is:
   A. contraindicated in patients with active HBV infection and associated GN.
   B. usually indicated in patients with active HBV infection and associated GN.

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